Information Systems Course Project:
Invoice Forecasting and Predictive Modeling System
Statement of Work and Analysis Report

Course Project Essay

Management Information Systems (MIS)

August 2026

Section 1: Statement of Work (SOW)

1.1 Basic Project Information

Project Description: The Invoice Forecasting and Predictive Modeling System is an enterprise information system designed to automate invoice trend forecasting, cash flow prediction, and financial data visualization through custom Python notebooks and an integrated web interface.

Project Sponsor: Chief Financial Officer (CFO) and Director of Financial Operations.

Team Members and Roles:

- Project Manager: Oversees timeline, resource allocation, and stakeholder communications.
- Lead Data Analyst: Engineers predictive models, handles data cleansing, and develops Python forecasting notebooks.
- Systems Architect: Designs database schemas, API integrations, and secure web hosting infrastructure.
- QA Specialist: Develops test cases, validates data integrity, and performs system verification.

1.2 Summary of the Business Need

The organization currently relies on manual, spreadsheet-based invoice tracking, which leads to significant delays, data siloing, and inaccurate cash flow forecasting. The business need is to establish an automated predictive tool that aggregates historical transaction records, predicts future invoice trends in real-time, and provides executive dashboards to support proactive financial planning and risk mitigation.

1.3 Key Business Requirements

- The business must be able to forecast monthly and quarterly cash inflows and invoice volumes with an accuracy rate of at least 85%.
- Financial analysts must be able to generate interactive visualization reports within seconds rather than hours of manual spreadsheet compilation.
- The business requires automated data validation protocols to flag unmapped customer IDs or missing invoice line items prior to financial reporting.

1.4 Business Value Estimates (Benefits)

- **Hard Tangible Benefits:** Direct reduction in administrative labor hours by 45%, saving an estimated $50,000 annually in manual data entry overhead.
- **Soft Tangible Benefits:** Faster month-end closing cycles and improved accuracy in working capital allocation.
- **Intangible Benefits:** Enhanced executive decision-making capabilities, improved audit readiness, and heightened employee morale through the elimination of repetitive manual reconciliation tasks.

1.5 Preliminary Work Breakdown Structure (WBS)

1. **Phase 1: Project Initiation & Planning**
    - 1.1 Stakeholder interviews and requirements sign-off
    - 1.2 Resource allocation and scheduling
2. **Phase 2: Analysis & System Modeling**
    - 2.1 Process modeling and Data Flow Diagram (DFD) creation
    - 2.2 Entity-Relationship Diagram (ERD) design

3. **Phase 3: Design & Development**
    - 3.1 Python predictive model engineering
    - 3.2 Web interface and dashboard development
4. **Phase 4: Testing & Deployment**
    - 4.1 System and unit testing via test cases
    - 4.2 User training and final rollout

Section 2: Analysis Report

2.1 System Requirements

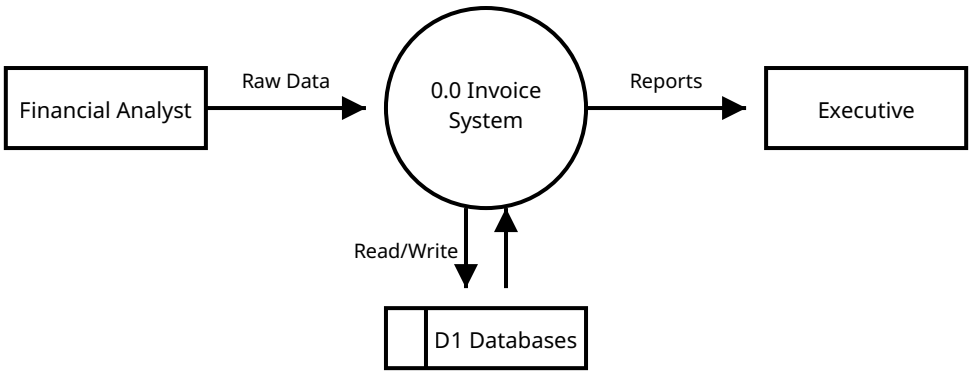
- **SR-01:** The system must accept CSV, Excel, and SQL database connectors for historical invoice ingestion.
- **SR-02:** The prediction engine must execute Python regression algorithms to generate 90-day rolling invoice forecasts within 5 seconds of query execution.
- **SR-03:** The user interface must enforce Role-Based Access Control (RBAC), restricting financial modification privileges to authorized accountants and data analysts.

2.2 Test Cases

- **Test Case 1 (TC-01: Invoice Data Ingestion):** Input a corrupted CSV file missing customer ID fields. *Expected Result:* The system rejects the file and generates an automated error log identifying the specific unmapped rows without crashing the application.
- **Test Case 2 (TC-02: Predictive Model Calculation):** Run a 90-day forecast with a standard historical dataset. *Expected Result:* The Python backend successfully computes projected cash flow and displays graphical trends on the executive dashboard within acceptable performance thresholds.
- **Test Case 3 (TC-03: Access Control Enforcement):** Attempt to modify financial parameters using a standard viewer account. *Expected Result:* The system blocks the action and displays an unauthorized access warning message.

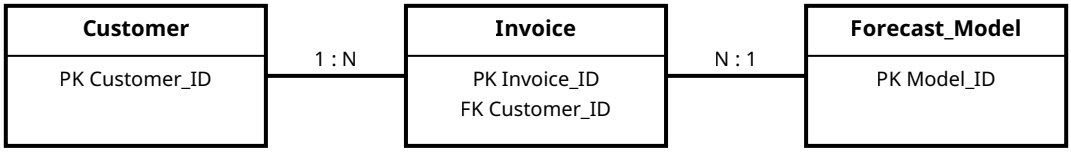
2.3 Data Flow Diagrams (DFDs) - Context & Level 0 Summary

The Data Flow Diagram models the movement of data within the system:



2.4 Entity-Relationship Diagrams (ERDs) - Data Model Summary

The ERD defines the relational data structure supporting the system:



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